

# Rational Curves on Compact Kähler Manifolds

Kähler MMP

Yi Li

The aims of this note are two:

- (1) We will study rational curves on Kähler varieties [HP16, CH20],
- (2) We will give an introduction to the BDPP theorem for projective and Kähler manifolds [BDPP13, Ou25].

Some further discussion and applications can be found in [Cone Theorem for Kähler MMP](#).

## Contents

---

|          |                                                                                                 |          |
|----------|-------------------------------------------------------------------------------------------------|----------|
| <b>1</b> | <b>Duality Between Various Positive Cones</b>                                                   | <b>2</b> |
| <b>2</b> | <b>BDPP Theorem for Projective manifolds</b>                                                    | <b>3</b> |
| <b>3</b> | <b>Ou's Proof of BDPP conjecture for Kähler manifolds</b>                                       | <b>5</b> |
| 3.1      | Algebraic integrability criteria under Kähler setting . . . . .                                 | 5        |
| 3.2      | Pseudo-effectiveness of the adjoint class . . . . .                                             | 5        |
| 3.3      | Relative Albanese reduction . . . . .                                                           | 5        |
| 3.4      | Proof of BDPP conjecture for compact Kähler manifold . . . . .                                  | 5        |
| <b>4</b> | <b>Several Applications of BDPP Theorem</b>                                                     | <b>5</b> |
| 4.1      | Applications of duality of pseudo-effective cone and cone of movable curves                     | 5        |
| 4.2      | Producing rational curves using BDPP conjecture . . . . .                                       | 5        |
| 4.3      | Cone theorem using BDPP conjecture . . . . .                                                    | 5        |
| <b>5</b> | <b>Höring–Peternell's approach of Mori bend-and-break for Kähler threefolds</b>                 | <b>6</b> |
| 5.1      | Mori bend-and-break for $K_S$ -negative curves on smooth projective uniruled surfaces . . . . . | 7        |
| 5.2      | Curves with large anti-canonical degree are deformable . . . . .                                | 7        |
| 5.3      | Proof of main theorem . . . . .                                                                 | 7        |

|          |                                                                                |          |
|----------|--------------------------------------------------------------------------------|----------|
| <b>6</b> | <b>Cao–Höring’s approach to producing rational curves for Kähler manifolds</b> | <b>7</b> |
| 6.1      | Pseudo-effectiveness of the adjoint class . . . . .                            | 8        |
| 6.2      | Subadjunction on the minimal log canonical center . . . . .                    | 8        |
| 6.3      | Proof of Cao–Höring’s main theorem . . . . .                                   | 9        |

## 1 Duality Between Various Positive Cones

In this section, we study the duality between various cones in projective and Kähler settings. The following theorem establishes the duality between the pseudo-effective cone and the movable cone on a projective manifold.

**Lemma 1.1.** *Let  $X$  be a projective manifold. Let  $\gamma \in N_1(X)$  be a movable class. Given any prime divisor  $E$ , there exists a representative  $\gamma_E$  such that  $\gamma_E$  intersects  $E$  properly and  $\gamma \equiv \gamma_E$ .*

**Remark 1.2.** It is unclear whether the result also holds for compact Kähler manifolds.

**Proof.** ■

**Theorem 1.3.** *Let  $X$  be a projective manifold. Then the pseudo-effective cone is dual to the cone of movable curves*

$$\mathcal{E} = \overline{\text{Mov}(X)}^\vee.$$

*In other words, a divisor is pseudo-effective if and only if it has non-negative intersection with every movable curve.*

**Remark 1.4.** David [?] proved . . .

**Remark 1.5.** Let us briefly sketch the idea of the proof.

**Proof.** Let  $C$  be a movable curve. By Lemma 1.1, we can choose  $C'$  such that  $C \equiv C'$  and  $C'$  meets the given pseudo-effective divisor properly. Hence

$$\mathcal{E} \subset \overline{\text{Mov}(X)}^\vee.$$

Conversely, if the inclusion is strict, then there exist

$$\xi \in \partial(\mathcal{E}(X)), \quad \xi \in \text{int}\left(\overline{\text{Mov}(X)}^*\right).$$

We want to deduce a contradiction. Since  $X$  is projective, we can find an ample divisor  $H$

such that  $\xi - \epsilon H$  is still in the movable cone. Thus

$$\frac{\xi \cdot C}{H \cdot C} \geq \epsilon, \quad \text{for every } C \in \overline{\text{Mov}}(X).$$

On the other hand we can apply Fujita approximation to the class  $\xi + tH$  for the ample  $H$ . This gives

$$\mu_t: X_t \rightarrow X$$

such that

$$\mu_t^*(\xi + tH) = A_t + E_t.$$

Choose  $C = \mu_{t*} A_t^{n-1}$ , then apply the asymptotic orthogonality of the Fujita approximation to  $\xi \cdot C$  and the Teissier–Khovanskii inequality to deduce an upper bound

$$\delta_t \geq \frac{\xi \cdot C}{H \cdot C} \geq \epsilon$$

with  $\delta_t \rightarrow 0$  as  $t \rightarrow 0$  (here  $\delta_t$  is a constant depending on  $\text{vol}(A_t)$ ; since  $\text{vol}(\xi) = 0$ , by Fujita approximation  $\text{vol}(A_t) \rightarrow 0$  as  $t \rightarrow 0$ ). ■

One can generalize the duality theorem to normal Moishezon spaces using a standard blow-up argument.

**Theorem 1.6.** *Let  $X$  be a normal Moishezon space. Then the pseudo-effective cone is dual to the movable cone of curves.*

**Proof.** ■

Using the duality theorem, we can show that the cone of nef curves coincides with the movable cone of curves.

**Theorem 1.7.** *Let  $X$  be a normal Moishezon space. Then the Batyrev nef cone coincides with the movable cone of curves.*

## 2 BDPP Theorem for Projective manifolds

The projective uniruled manifold is characterized by the pseudo-effectiveness of the canonical bundle.

**Lemma 2.1.** *Given a movable curve  $C$ , there exists a covering family  $\bigcup_{t \in S} C_t$  containing  $C$  that covers a dense open subset of  $X$ . To be more precise, we can find a diagram*

$$\begin{array}{ccc} C & \xrightarrow{\phi} & X \\ f \downarrow & & \\ S & & \end{array}$$

where  $f$  is a fibration with fibers  $C_t$ ,  $\phi$  is a dominant generically finite morphism, and  $\{C_t\}_{t \in S}$  lie in the same numerical class.

**Proof.** ■

**Theorem 2.2** ([BDPP13, Corollary 0.3]). *Let  $X$  be a projective manifold. Then  $X$  is uniruled if and only if  $K_X$  is not pseudo-effective.*

**Remark 2.3.** One direction of the proof is easy, and can be adapted to the Kähler manifold. The converse direction (that  $K_X$  is not pseudo-effective implies  $X$  is uniruled) is non-trivial, which requires the Mori bend-and-break technique and the duality between the pseudo-effective cone and the movable cone.

**Remark 2.4.** Miyaoka and Mori [?] proved that a projective manifold is uniruled if and only if there exists an open subset through every point of which there passes a  $K_X$ -negative curve. For more discussion about the Miyaoka–Mori theorem (and various properties of uniruled manifolds) see Note 15.

**Proof.** It suffices to prove that if  $K_X$  is not pseudo-effective, then  $X$  is uniruled. By the duality of the pseudo-effective cone and the movable cone, there exists a movable curve  $C$  such that

$$K_X \cdot C < 0.$$

By Lemma 2.1, we can produce a covering family of  $K_X$ -negative irreducible curves using the movable curve  $C$ . ■

We can generalize the BDPP theorem to the singular case.

**Theorem 2.5.** *Let  $(X, B)$  be a  $\mathbb{Q}$ -factorial log pair. If  $K_X + B$  is not pseudo-effective, then  $X$  is uniruled.*

**Remark 2.6.** Rational curves on singular spaces are tricky. See Note 9 for further discussion.

**Proof.** Take a log resolution

$$f: X' \rightarrow X$$

such that  $f^*(K_X + B) = K_{X'} + B'$ . Since being uniruled is a birational invariant, if  $X$  is not

uniruled, then neither is  $X'$ . Then by the BDPP theorem proved above,  $K_{X'}$  is pseudo-effective, thus  $K_X$  is pseudo-effective. Since  $B$  is effective,  $K_X + B$  is pseudo-effective. ■

We can also show the converse direction for canonical singularity.

### Theorem 2.7.

The following example indicates that the BDPP theorem may fail for singular varieties.

We can characterize the uniruled variety using a subsheaf of the tangent sheaf.

**Theorem 2.8.** *Let  $X$  be a projective manifold and let  $\mathcal{F} \subset T_X$  be a coherent subsheaf such that  $\det(\mathcal{F}^*)$  is not pseudo-effective. Then  $X$  is uniruled.*

**Proof.** ■

## 3 Ou's Proof of BDPP conjecture for Kähler manifolds

Recently, Ou [Ou25] proved the BDPP conjecture for compact Kähler manifolds. In this section, we briefly introduce the result that he proved.

### 3.1 Algebraic integrability criteria under Kähler setting

The

### 3.2 Pseudo-effectiveness of the adjoint class

### 3.3 Relative Albanese reduction

### 3.4 Proof of BDPP conjecture for compact Kähler manifold

## 4 Several Applications of BDPP Theorem

### 4.1 Applications of duality of pseudo-effective cone and cone of movable curves

### 4.2 Producing rational curves using BDPP conjecture

### 4.3 Cone theorem using BDPP conjecture

Hörling–Peternell [HP24] proved the following cone theorem for klt Kähler pairs. We will discuss the details of the proof in [REFERENCE].

**Theorem 4.1.**

## 5 Höring–Peternell's approach of Mori bend-and-break for Kähler threefolds

In this section, we briefly summarize the approach of Höring–Peternell [HP16] on Mori's bend-and-break for Kähler threefolds.

Let us first define the working condition for this section.

**Definition 5.1 (Condition (\*)).** Let  $X$  be a  $\mathbb{Q}$ -factorial normal compact Kähler threefold with at most terminal singularity, assume  $K_X$  is pseudo-effective, such that  $S_1, \dots, S_k$  are the components of the negative part of Zariski decomposition of  $K_X$ .

As in the standard Mori bend-and-break, if the anti-canonical degree is greater than a threshold  $b$ , then the curve breaks and produces a rational curve. Let us first define the threshold for our theorem

$$b = \max\{1, -K_X \cdot Z \mid Z \in S_{j,\text{Sing}} \text{ or } Z \in S_j \cap S_i\}.$$

**Theorem 5.2.** *Under condition (\*)*

1. *If  $C$  is a curve with large anti-canonical degree (say  $-K_X \cdot C > b$ ), then it breaks into*

$$[C] = [C_1] + [C_2].$$

2. *Let  $\overline{\text{NE}}(X)$  have a  $K_X$ -negative extremal ray  $\mathbb{R}_+[\Gamma]$ , such that the representative  $\Gamma$  is not very rigid. Then we can find a representative  $C \in \mathbb{R}_+[\Gamma]$  such that  $\dim_C \text{Chow}(X) > 0$ , and  $\mathbb{R}_+[\Gamma]$  contains rational curves.*

**Remark 5.3.** Let us first briefly sketch the idea of the proof: The general idea is that if the anti-canonical degree is large, then the curve  $C$  is deformable in the sense that  $\dim_C \text{Chow}(X) > 0$ .

We then prove that the deformation of the curve is contained in a component  $S_i$  of the negative part of the Zariski decomposition  $N(K_X)$ . (Note that the surface  $S_i$  has  $K_{S_i}$  not pseudo-effective, thus it is a uniruled surface.)

Therefore, we reduce the problem to the surface  $S_i$  and prove that a  $K_{S_i}$ -negative curve on the uniruled surface breaks and produces a rational curve.

There are several issues in the idea above.

Before proceeding to the proof, let us introduce an important result that is repeatedly used: the minimal resolution of the surface decreases the  $K_X$ -degree.

## 5.1 Mori bend-and-break for $K_S$ -negative curves on smooth projective uniruled surfaces

**Lemma 5.4.** *Let  $S$  be a smooth projective uniruled surface. Let  $C$  be a curve on  $S$  such that*

1. *(Existence of rational curves) If  $K_S \cdot C < 0$ , then  $C$  is numerically equivalent to  $\sum_i c_i [C_i]$  for some rational curves  $C_i$ .*
2. *(Curve with large anti-canonical degree breaks) If  $K_S \cdot C < -4$ , then there exists a 1-cycle  $\sum_{i=1}^m C_i$  with  $m \geq 2$  components such that*

$$[C] = \left[ \sum_{i=1}^m C_i \right],$$

*with  $K_S \cdot C_1 < 0$  and  $K_S \cdot C_2 < 0$ .*

| **Proof.** ■

## 5.2 Curves with large anti-canonical degree are deformable

The technical core of the proof is the following lemma, which shows that a curve with large anti-canonical degree is deformable in the sense that  $\dim \text{Chow}_C(X) > 0$ .

**Lemma 5.5.** *Under the setting of (\*), assume that*

$$-K_X \cdot C > b,$$

*then  $\dim \text{Chow}_C(X) > 0$ .*

| **Proof.** ■

## 5.3 Proof of main theorem

We can now prove the main theorem.

| **Proof of Theorem 5.2.** ■

# 6 Cao–Höring’s approach to producing rational curves for Kähler manifolds

Cao–Höring found that the BDPP conjecture is useful for producing  $K_X$ -negative rational curves on a compact Kähler manifold when  $K_X$  is pseudo-effective but not nef. The main reference for this section is [CH20]. The technical core (and the most difficult part) of the proof is the subadjunction formula on the mlc center for compact Kähler manifolds.

**Theorem 6.1.** *Let  $X$  be a compact Kähler manifold such that  $K_X$  is pseudo-effective but not nef. Then there exists a  $K_X$ -negative rational curve  $C$  on  $X$ .*

The proof contains two essential ingredients:

- (a) The subadjunction formula on the mlc center on compact Kähler manifolds,
- (b) Pseudo-effectiveness of the adjoint class in a family.

## 6.1 Pseudo-effectiveness of the adjoint class

The main technical tool used in the proof of Cao–Höring’s theorem is the following pseudo-effectiveness theorem.

**Theorem 6.2** ([CH20, Theorem 5.2]). *Let  $X$  and  $Y$  be two compact Kähler manifolds and let  $f : X \rightarrow Y$  be a surjective map with connected fibres such that the general fibre  $F$  is simply connected and*

$$H^0(F, \Omega_F^2) = 0.$$

*Let  $\omega$  be a Kähler form on  $X$  such that  $c_1(K_F) + [\omega|_F]$  is a pseudoeffective class. Then  $c_1(K_{X/Y}) + [\omega]$  is pseudoeffective (absolute pseudo-effectiveness).*

**💡 Idea of proof.** The structure of the proof is standard. We first restrict our attention to a Stein open subset  $U_i \subset Y$ .

**Step 1.** Find a holomorphic line bundle  $L_i$  approximating the closed real  $(1, 1)$ -form  $\alpha + \omega$ .

**Step 2.** Construct a sequence of Bergman kernel metrics  $\varphi_{i,m}$  on the holomorphic line bundles  $L_m$ .

**Step 3.** Show that the limiting metric

$$\varphi_i := \lim_{k \rightarrow +\infty} \left( \sup_{m \geq k} \varphi_{i,m} \right)^*$$

exists and is a quasi-psh function.

**Step 4.** The metrics glue together on  $f^{-1}(Y_0)$  to yield a metric on  $c_1(K_{X/Y}) + [\omega]$ .

**Step 5.** The metric can be extended to all of  $Y$ .

## 6.2 Subadjunction on the minimal log canonical center

In this section, we only briefly review the result; the proof can be found in [Subadjunction On Kähler varieties](#). Using a Bergman-type metric, Cao–Höring proved the following subadjunction formula on the mlc center.

**Theorem 6.3.** *Let  $X$  be a compact Kähler manifold, and let  $\alpha$  be a cohomology class on  $X$  that is a modified Kähler class (cf. Definition 4.1). Suppose that  $Z \subset X$  is a maximal lc centre of the pair  $(X, \alpha)$ , and let  $v : \tilde{Z} \rightarrow Z$  be the normalisation. Then*

$$K_{\tilde{Z}} \cdot \omega_1 \cdot \dots \cdot \omega_{\dim Z - 1} \leq v^*(K_X + \alpha)|_Z \cdot \omega_1 \cdot \dots \cdot \omega_{\dim Z - 1},$$

where  $\omega_1, \dots, \omega_{\dim Z - 1}$  are arbitrary nef classes on  $\tilde{Z}$ .

### 6.3 Proof of Cao–Höring's main theorem

**Proof.** We divide the proof into two parts.

(I) Find a uniruled component  $Z$  in the null locus (which is guaranteed by the BDPP conjecture).

(II) Find a  $K_X$ -negative extremal ray on the uniruled component  $Z$  (which requires the subadjunction formula given in the previous section).

■

## References

- [BDPP13] S. Boucksom, J.-P. Demailly, M. Păun, and T. Peternell, *The pseudo-effective cone of a compact Kähler manifold and varieties of negative Kodaira dimension*, J. Algebraic Geom. **22** (2013), no. 2, 201–248.
- [CH20] Junyan. Cao and A. Höring, *Rational curves on compact Kähler manifolds*, J. Differential Geom. **114** (2020), no. 1, 1–39.
- [HP16] A. Höring and T. Peternell, *Minimal models for Kähler threefolds*, Invent. Math. **203** (2016), no. 1, 217–264.
- [HP24] C.D. Hacon and M. Păun, *On the canonical bundle formula and adjunction for generalized kähler pairs*, 2024.
- [Ou25] W.-H. Ou, *A characterization of uniruled compact kähler manifolds*, 2025.